Accelerating Mathematical Discovery

A Crash Course in AI and Computation

October 28 – November 1, 2026

Fairfield Inn & Suites Santa Cruz – Capitola

Welcome

In May we had the idea to host a conference around AI and Mathematics. There are two essential goals:

Welcome

In May we had the idea to host a conference around AI and Mathematics. There are two essential goals:

Goal #1

To showcase and encourage the adoption of computation and AI tools (through coding, formalization, verification, and other workflows).

Welcome

In May we had the idea to host a conference around AI and Mathematics. There are two essential goals:

Goal #1

To showcase and encourage the adoption of computation and AI tools (through coding, formalization, verification, and other workflows).

Goal #2

To identify areas of mathematics that can be aided with these tools, and to develop open-source solutions for this purpose.

Working Groups

This workshop is experimental, with time devoted for brainstorming.

Working Groups

This workshop is experimental, with time devoted for brainstorming.

- Is there a tool that could help you in your research?
  (E.g. a database of objects, a package in a CAS like Macaulay2, a program that formalizes your paper as you write it)

Working Groups

This workshop is experimental, with time devoted for brainstorming.

- Is there a tool that could help you in your research?
  (E.g. a database of objects, a package in a CAS like Macaulay2, a program that formalizes your paper as you write it)
- What will mathematics look like in the future, and how can we shape it?

Working Groups

This workshop is experimental, with time devoted for brainstorming.

- Is there a tool that could help you in your research?
  (E.g. a database of objects, a package in a CAS like Macaulay2, a program that formalizes your paper as you write it)
- What will mathematics look like in the future, and how can we shape it?
- Really, shoot for the stars.

Working Groups

This workshop is experimental, with time devoted for brainstorming.

- Is there a tool that could help you in your research?
  (E.g. a database of objects, a package in a CAS like Macaulay2, a program that formalizes your paper as you write it)
- What will mathematics look like in the future, and how can we shape it?
- Really, shoot for the stars.

On Friday we'll get into groups with concrete problems to work on.

Working Groups

This workshop is experimental, with time devoted for brainstorming.

- Is there a tool that could help you in your research?
  (E.g. a database of objects, a package in a CAS like Macaulay2, a program that formalizes your paper as you write it)
- What will mathematics look like in the future, and how can we shape it?
- Really, shoot for the stars.

On Friday we'll get into groups with concrete problems to work on.

Then Saturday and Sunday we will work on developing open-source solutions for the problems we've identified.

Working Groups

This workshop is experimental, with time devoted for brainstorming.

- Is there a tool that could help you in your research?
  (E.g. a database of objects, a package in a CAS like Macaulay2, a program that formalizes your paper as you write it)
- What will mathematics look like in the future, and how can we shape it?
- Really, shoot for the stars.

On Friday we'll get into groups with concrete problems to work on.

Then Saturday and Sunday we will work on developing open-source solutions for the problems we've identified.

We'll use **Git** to manage version-control for these projects, and they will be linked to in the repo:
`https://github.com/aimathconference/accelerating-math-discovery.git`.

AI Inference and Other Tools

Thanks to support from Renaissance Philanthropy's AI for Math Fund, we're able to provide the following tools:

1. Claude and Claude Code

- **One year** subscription to Claude for funded participants (check your registration email if you haven't already joined)

AI Inference and Other Tools

Thanks to support from Renaissance Philanthropy's AI for Math Fund, we're able to provide the following tools:

1. Claude and Claude Code

- **One year** subscription to Claude for funded participants (check your registration email if you haven't already joined)
- If you haven't already installed Claude Code, there will be a chance later today

AI Inference and Other Tools

Thanks to support from Renaissance Philanthropy's AI for Math Fund, we're able to provide the following tools:

2. Google Colab Pro+

- **One month** subscription to Google Colab Pro+ ($49.99/month)

AI Inference and Other Tools

Thanks to support from Renaissance Philanthropy's AI for Math Fund, we're able to provide the following tools:

2. Google Colab Pro+

- **One month** subscription to Google Colab Pro+ ($49.99/month)
- Access to cloud compute (up to 51 GB RAM) and direct access to running on a GPU (RTX PRO 6000, H100, A100 among others) or a TPU (v5e-1, v6e-1)

AI Inference and Other Tools

Thanks to support from Renaissance Philanthropy's AI for Math Fund, we're able to provide the following tools:

2. Google Colab Pro+

- **One month** subscription to Google Colab Pro+ ($49.99/month)
- Access to cloud compute (up to 51 GB RAM) and direct access to running on a GPU (RTX PRO 6000, H100, A100 among others) or a TPU (v5e-1, v6e-1)
- Reimbursement must be submitted no later than 11/15/26

AI Inference and Other Tools

Thanks to support from Renaissance Philanthropy's AI for Math Fund, we're able to provide the following tools:

3. Nvidia DGX Spark

- Local API access to an Nvidia DGX Spark (connect through Tailscale link sent this morning)

AI Inference and Other Tools

Thanks to support from Renaissance Philanthropy's AI for Math Fund, we're able to provide the following tools:

3. Nvidia DGX Spark

- Local API access to an Nvidia DGX Spark (connect through Tailscale link sent this morning)
- Running Nvidia's Qwen3.8-Flash-Next-FP4-Local on vLLM on port 8000, Open-WebUI on port 3000

AI Inference and Other Tools

Thanks to support from Renaissance Philanthropy's AI for Math Fund, we're able to provide the following tools:

3. Nvidia DGX Spark

- Local API access to an Nvidia DGX Spark (connect through Tailscale link sent this morning)
- Running Nvidia's Qwen3.8-Flash-Next-FP4-Local on vLLM on port 8000, Open-WebUI on port 3000
- 8 requests can run concurrently, queued otherwise

Other logistics

Near the end of the week, you will receive an email for how to submit reimbursement for your travel/lodging.

Other logistics

Near the end of the week, you will receive an email for how to submit reimbursement for your travel/lodging.

Any questions?

Thanks for coming

hope you enjoy :)